

Execution Plan for Portugal Property Intelligence Infrastructure

PropCheck · InspectOS · REZERVA · AIRCS

1. Strategic Objective

The objective is to build the **foundational infrastructure for a property intelligence platform in Portugal**.

The platform will integrate multiple types of property data to create a **standardized property risk evaluation system**.

The ecosystem will operate through four main components:

- **InspectOS** — inspection infrastructure
- **AIRCS** — property intelligence engine
- **PropCheck** — buyer intelligence platform
- **REZERVA** — seller capture platform

These systems together create a **complete property intelligence ecosystem**.

2. System Architecture Overview

The entire system operates through a **data-driven architecture** where multiple property data sources feed the AIRCS intelligence engine.

The results are then delivered to the **buyer platform (PropCheck)** and the **seller platform (REZERVA)**.

Each completed transaction generates new inspection data, which feeds back into the system and improves the models.

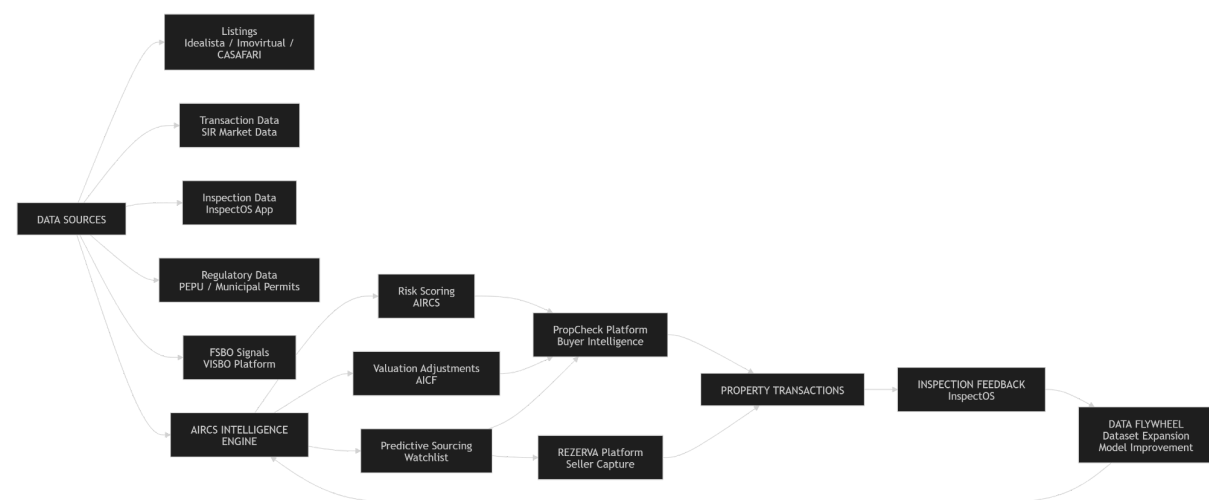
The ecosystem starts with multiple data sources including listings, inspections, regulatory datasets, and FSBO signals.

These data streams feed the AIRCS intelligence engine, which produces property risk scoring, valuation corrections, and predictive sourcing signals.

The outputs power two platforms: PropCheck for buyer intelligence and REZerva for seller capture.

Transactions generate inspection feedback that feeds back into the system, creating a data flywheel that continuously improves the accuracy of the AIRCS model.

Core System Flow



3. Property Data Sources

The system integrates several property intelligence datasets.

Listing Platforms

- Idealista
- Imovirtual
- CASAFARI

Transaction Data

- SIR property transaction dataset

Inspection Data

- InspectOS mobile inspection application

Regulatory Data

- PEPU permit database
- municipal records

FSBO Data Signals

- VISBO seller platform

These datasets are combined to create a **multi-layer property intelligence dataset**.

4. AIRCS Intelligence Engine

The **Asset Integrity & Regulatory Compliance Score (AIRCS)** is the analytical core of the system.

AIRCS converts complex property information into a **single standardized risk score (0–100)**.

AIRCS Risk Dimensions

Dimension	Weight
Physical Integrity	30%
Regulatory Compliance	25%
Market Alignment	20%
Energy Performance	15%
Environmental Risk	10%

The scoring system enables buyers and investors to **evaluate property risk quickly and objectively**.

5. AICF Valuation Adjustment

The **Asset Integrity Correction Factor (AICF)** adjusts property value using inspection findings.

Formula

Adjusted Value = Market Comparable Value – Estimated Renovation Costs

Example

Market value estimate: €490,000

Renovation cost: €55,000

Adjusted property value:

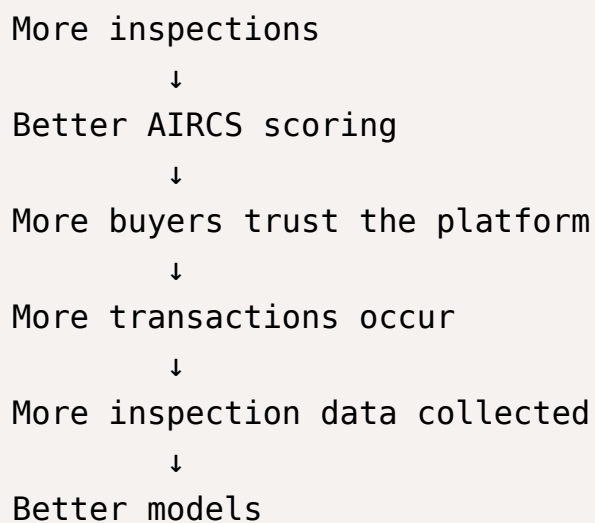
€435,000

This method provides **realistic financial expectations for property buyers**.

6. Data Flywheel (Strategic Advantage)

One of the key advantages of the platform is the **data flywheel effect**.

The process works as follows:



Over time this creates **a powerful proprietary dataset that competitors cannot easily replicate**.

7. Execution Plan (90-Day Implementation)

The implementation follows the **execution roadmap defined in the Master Strategy Report**.

Phase 1 — Platform Foundations (Day 0–30)

Objectives

Build the **data infrastructure and inspection pilot**.

Key Tasks

- Build **data ingestion pipelines**
- Integrate **listing platforms**
- Integrate **PEPU permit data**
- Deploy **InspectOS inspection template**
- Complete **10 pilot inspections**

Technical Work

Data infrastructure setup:

- property database (PostgreSQL + PostGIS)
- lakehouse architecture (Bronze / Silver / Gold layers)
- web scraping pipelines
- inspection data collection system

Expected Deliverables

- operational data pipeline
 - 10 inspected properties
 - initial inspection dataset
 - prototype AIRCS data features
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Phase 2 — Intelligence Model Development (Day 31–60)

Objectives

Develop the **AIRCS scoring model and valuation adjustments**.

Key Tasks

- build AIRCS scoring model
- develop AICF valuation adjustments
- evaluate **50 properties**

Model Development

Inspection and market data will be processed through the **AIRCS data pipeline**:

1. data ingestion
2. data cleaning
3. feature engineering
4. risk scoring
5. model calibration

Expected Deliverables

- AIRCS scoring engine (MVP)
 - AICF valuation system
 - 50 evaluated properties
 - initial predictive data signals
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Phase 3 — Predictive Intelligence & Institutional Pilot (Day 61–90)

Objectives

Introduce **predictive sourcing and institutional integration**.

Key Tasks

- launch **Predictive Property Watchlist**
- pilot **AIRCS API with bank**
- begin **SIR licensing negotiation**

Predictive Signals

Properties likely to enter the market may be identified using:

- inheritance filings
- IMI tax irregularities
- energy certificate expiration
- permit activity
- long ownership duration

Expected Deliverables

- predictive sourcing system
 - AIRCS API prototype
 - institutional pilot discussion
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Phase 4 — Platform Expansion (Day 91–180)

Objectives

Scale the ecosystem and launch commercial operations.

Key Tasks

- expand inspections to **200+ properties**
- launch **PropCheck subscription**
- begin **commercial pilots**

This phase transitions the platform from **prototype to operational product**.

8. Early Platform Launch Strategy

Initially, the platform will operate using **semi-manual services**.

Examples of early services:

- property risk reports

- property price evaluation reports
- bureaucracy checks
- due diligence reports

Users will see **basic analytics on the website**, and they can request a **full report for a fee**.

This approach allows the platform to:

- generate early revenue
- validate demand
- collect valuable data.

DATA PIPELINE—STEP-BY-STEP

1. **Property canonicalization service** — central index keyed by cadastral references/address; collect unique identifiers.
2. **Bronze ingestion** — raw JSON/CSV blobs from scrapers/APIs + inspection raw outputs.
3. **Normalization (Silver)** — standardize fields (address, coordinates, owner anonymization tokens, permit dates, EPC codes).
4. **Feature engineering (Gold)** — compute physical integrity score features, permit compliance flags, price deviation, EPC aging, and environmental risk indicators.
5. **AIRCS scoring** — apply weights to normalized indicators and return score + component breakdown.
6. **AICF calculation** — aggregate estimated renovation costs into a single correction factor and produce adjusted valuation fields.
7. **Monitoring & lineage** — dataset versioning and logging per dataset version.

Immediate engineering tasks (first meeting asks):

- Create a backlog ticket list for each pipeline stage with acceptance criteria.
- Provide a minimal schema of the Gold-layer feature table (fields + types) to the team today.

PILOT DESIGN (InspectOS + AIRCS MVP)

Pilot scope: 10–50 properties in Lisbon/Cascais region, mix of apartments and small houses, varied ages.

Pilot steps:

1. Define inspection template + photo checklist + contractor cost estimate fields.
2. Recruit 2–4 local inspectors or contractors (freelance) and train them on InspectOS v1.
3. Run inspections, collect reports, upload to Bronze.
4. Normalize reports → compute physical integrity indicators.
5. Generate AIRCS and AICF reports.
6. Deliver PropCheck report to buyer test users / internal reviewers and collect feedback.
7. Iterate template and cost model.

Pilot success metrics:

- Completed inspections / week.
- Time-to-report (target <72 hours in semi-manual flow).
- Buyer feedback NPS on report usefulness (>7/10 target).
- Correlation checks between AICF and actual contractor estimates (error <20% target after calibration).

PROGRAMMATIC SEO & VISBO (quick launch sequence)

- Day 0–7: finalize content schema for Bureaucracy Scanner and municipal pages.
- Day 8–21: scrape municipal datasets / compile bureaucratic steps per municipality.
- Day 15: launch 50 SEO pages seeded with data + tools (bureaucracy scanner MVP).

- Ongoing: add VISBO FSBO intake forms on SEO pages to capture seller leads & early signals.

Lead: Helios (VISBO + SEO), with Erdal feeding scraped data.

SECURITY, GDPR, AND DATA GOVERNANCE (must-decide items in meeting)

- Decide data retention policy and PII handling: strip personal identifiers from analytic datasets; store identifiers only in secure vault with access rules.
- Consent flows for inspections & VISBO onboarding.
- Encryption at rest and transit.
- Audit logging and dataset versioning.

Ask in meeting: **Does supervisor approve creating a privacy & security checklist and a minimal budget for secure storage/hosting?**

9. Data Pipeline Development

A **structured data pipeline** is essential for the platform.

Pipeline Stages

1. Data ingestion
2. Data cleaning and normalization
3. Feature engineering
4. AIRCS risk scoring
5. Model calibration

This architecture ensures the platform becomes **more accurate as the dataset grows**.

10. Data Acquisition Strategy

The system will use multiple data acquisition methods:

- APIs
- public datasets
- web scraping
- proprietary data collection

This strategy prevents dependency on a **single external data provider**.

11. Security and Data Governance

Because the platform processes sensitive information, strong security measures are required.

Key Principles

- GDPR compliance
- encryption of sensitive data
- secure data storage
- strict access control
- separation of personal and analytical datasets

Proper security architecture must be implemented **from the beginning**.

12. Team Roles

Ash

- **Technical architecture advisor** responsible for designing and validating the overall system architecture.
- Provide **guidance on the development of the property data pipeline**, including data ingestion, structuring, and feature preparation.
- Contribute to the **development and refinement of the AIRCS scoring model**.
- **Validate technical implementation and system design** to ensure alignment with the strategy report.

Team Member 1

- data scraping and ingestion pipelines

Team Member 2

- data structuring and dataset preparation

Supervisor

- strategic guidance
- institutional partnerships
- licensing negotiations

12.1 Technical Advisory Role (Ash)

My role in the project is primarily **technical advisory and architectural guidance**, rather than direct implementation of every technical task.

His responsibilities include:

System Architecture Design

Design and validate the **overall platform architecture**, including:

- data infrastructure
- AIRCS intelligence engine
- property data pipeline
- platform ecosystem structure.

Data Pipeline Guidance

Provide technical guidance for the development of the **property data pipeline**, including:

- data ingestion strategies
- dataset structuring
- feature engineering for AIRCS models.

AIRCS Model Development

Actively contribute to the **development and refinement of the AIRCS scoring model**, including:

- defining the scoring framework and weighting logic
- designing feature inputs used in the AIRCS model
- improving model calibration as inspection and transaction data become available
- validating risk scoring outputs.

Technical Validation

Review and validate the work produced by the data engineering team to ensure that:

- the system architecture follows the strategy report
- datasets are structured correctly
- analytical models are implemented properly.

Strategic Technical Decisions

Assist the team in making technical decisions regarding:

- data infrastructure
- system scalability
- model development.

Final Team Structure

You can present the team structure like this:

Role	Responsibility
Achref	Technical architecture advisor
Team Member 1	Data ingestion and scraping pipelines
Team Member 2	Dataset structuring and preparation
Pieter (Supervisor)	Strategic direction and partnerships

My role will focus mainly on designing the technical architecture, guiding the data pipeline structure, and validating the system implementation, while Erdal and Helios will focus on the data engineering and collection tasks.

13. Immediate Next Steps

Short-term actions include:

- prepare **data pipeline architecture**
 - identify **initial data sources**
 - deploy **inspection templates**
 - begin **pilot inspections**
 - organize **team data collection tasks**
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